

Done in a hurry so that it is guaranteed to have typos and errors. I will include the solutions to the other problems later.

4.1) A clear way to think about this problem is to view it as a one-dimensional problem and to imagine the neutrons with different spins entering a region with the magnetic field thus experiencing a different potential in that region. There will be neutrons that are reflected and transmitted and one can compute them. We need only use energy conservation.

Recall that in a one-dimensional problem in a region with a potential the wave vector or the momentum is obtained by conservation of the energy. The magnetic field interaction is described by the Hamiltonian $-g_n\mu_NBS_z/\hbar$. The neutron g -factor is denoted by g_n and the nuclear (Bohr) magneton by μ_N . Thus the momentum in the region of the field $p_{\pm}$ for the up and down spins can be written in terms of the initial momentum p_0 as

$$\frac{p_0^2}{2M} = \frac{p_{\pm}^2}{2M} \mp \frac{g_n\mu_NB}{2}.$$

Thus we obtain

$$p_{\pm} = p_0 \left(1 \pm \frac{g_n\mu_NBM}{p_0^2} \right)^{1/2} \approx p_0 \pm \frac{g_n\mu_NBM}{2p_0}.$$

Think of a free-particle wave function $e^{ipx/\hbar}$ with different momenta for the two spins. This is a bit like a double refraction in classical optics. In the absence of the field the neutron wave function will be multiplied by a phase factor $e^{i\frac{p_0\ell}{\hbar}}$ where ℓ denotes the length of the region over which the field acts. The two different orientations of the spin will induce an extra phase shift given by

$$\pm \frac{g_n\mu_NBM\ell}{2p_0\hbar} = \pm \frac{g_n\mu_NBM\lambda\ell}{2p_0h\hbar}$$

where we have used the de Broglie relation. This yields the given equation. Grumble, grumble. Off by a factor of 2. OK, they give $g_n = -1.91$ and in my convention $g_n = 2 \times (-1.91)$. Phew! Please substitute the numbers and admire the figure.

4.2) Momentum and angular momentum conservation determine the direction of the momentum of the pion to be along the $-z$ axis and the spin of the proton to be up along the z -axis respectively.

(b) By angular momentum conservation the spin of the proton is along $-z$.

(c1) The amplitude for the lambda to be up along the z' axis given that it is up along the z -axis is $\cos(\theta/2)$ from the book and discussion in class. So the amplitude for the proton to emerge up along z' is $a \cos(\theta/2)$.

(c2) The amplitude for the lambda to be down along the z' axis given that it is up along the z -axis is $\sin(\theta/2)$ from the book and discussion in class. So the amplitude for the proton to emerge up along z' is $b \sin(\theta/2)$.

(d) The probability for the proton to come out along z' is the sum of the two probabilities and is given by $|a|^2 \cos^2(\theta/2) + |b|^2 \sin^2(\theta/2)$. Here is the key quantum mechanical principle. The two processes (proton with spin up and down along z') are distinguishable and therefore, we add the probabilities and not the amplitudes. Please see Feynman Vol. III where he emphasizes this clearly. This can be written as

$$\frac{|a|^2 + |b|^2}{2} + \left(\frac{|a|^2 - |b|^2}{2} \right) \cos \theta.$$

(e) Under mirror reflection about the x -axis (note that this is not the full parity operation) $p_z \rightarrow p_z$ and the angular momentum along the z -direction $x p_y - y p_x$ changes sign since x does and p_y does not. Therefore we would expect the probabilities for the two processes to be the same: $|a|^2 = |b|^2$.

(f) The fact that the experiment shows that the coefficient of the $\cos \theta$ term is non-zero shows that parity is violated in this process. This was one of the early tests of the behavior of weak interactions. We can use the data (of the form $(1 + \gamma \cos \theta)$) to determine $\frac{|a|^2 - |b|^2}{|a|^2 + |b|^2}$ i.e., $|b|/|a|$. Experimentally $\gamma \approx -3/5$ yielding $|b| \approx 2|a|$. It is not so easy to determine the magnitudes of the individual amplitudes without more work. Still pretty good given that you do not know anything about weak interactions.

4.3 For the electron the intrinsic magnetic moment is $-g \frac{e}{2m} \vec{S}$. Thus in a magnetic field $\vec{B}$ the Hamiltonian is $-\vec{\mu} \cdot \vec{B} = \frac{e}{m} \vec{B} \cdot \vec{S}$.

The Hamiltonian for this problem is $\frac{eB}{m} S_x$. For convenience we write this as ωS_x where $\omega \equiv eB/m$. Since S_x is an angular momentum this has dimensions of $\hbar\omega$.

The Heisenberg equation of motion for the spin operator is

$$\frac{d\vec{S}(t)}{dt} = \frac{i}{\hbar} [H, \vec{S}(t)] = \frac{i\omega}{\hbar} [S_x(t), \vec{S}(t)].$$

I have left the time dependence explicit to remind you that these are Heisenberg operators. Thus we obtain in detail

$$\begin{aligned}\frac{dS_x}{dt} &= 0. \\ \frac{dS_y(t)}{dt} &= \frac{i\omega}{\hbar} [S_x(t), S_y(t)] = -\omega S_z(t). \\ \frac{dS_z(t)}{dt} &= \frac{i\omega}{\hbar} [S_x(t), S_z(t)] = \omega S_y(t).\end{aligned}$$

The solutions can be written down by inspection by analogy with classical equations of motion for a precessing spin:

$$\begin{aligned}S_x(t) &= S_x(0). \\ S_y(t) &= S_y(0) \cos(\omega t) - S_z(0) \sin(\omega t). \\ S_z(t) &= S_y(0) \sin(\omega t) + S_z(0) \cos(\omega t).\end{aligned}$$

You can check this by substitution into the equations of motion.

We wish to determine $[\Delta S_z(t)]^2$ for the state $|z+\rangle$. We need $\langle z+ | S_z^2(t) | z+\rangle$ and $\langle z+ | S_z(t) | z+\rangle$. The first is trivial since $S_z^2 = \frac{\hbar^2}{4} I$ and so we find $\frac{\hbar^2}{4}$. For the second we have

$$\langle z+ | S_z(t) | z+\rangle = \sin(\omega t) \langle z+ | S_y | z+\rangle + \cos(\omega t) \langle z+ | S_z | z+\rangle = \frac{\hbar}{2} \cos(\omega t).$$

We obtain

$$[\Delta S_z(t)]^2 = \frac{\hbar^2}{4} \sin^2(\omega t).$$

4.4 Writing $A\vec{S}_1 \cdot \vec{S}_2 = \frac{A}{2} (\vec{S}_1 + \vec{S}_2)^2 - \vec{S}_1 \cdot \vec{S}_1 - \vec{S}_2 \cdot \vec{S}_2$ and noting that $\vec{S}_1^2 = \vec{S}_2^2 = \frac{3}{4}\hbar^2 I$ and $\vec{S}^2$ where $\vec{S} = \vec{S}_1 + \vec{S}_2$ has eigenvalues $2\hbar^2$ and 0 for the triplet with $S = 1$ and singlet with $S = 0$ the result in Equation 15.1.23 follows.

We have $\vec{\mu}_e = -\frac{g_e e}{2m_e c} \vec{S}_e$ and $\vec{\mu}_p = \frac{g_p e}{2M c} \vec{S}_p$ we have

$$H_{hf} = -\frac{\vec{\mu}_e \cdot \vec{\mu}_p}{a_0^3} = A \vec{S}_e \cdot \vec{S}_p$$

we obtain the identification

$$A = \frac{2e}{2mc} \frac{5.6e}{2Mc} \frac{1}{a_0^3}.$$

Thus using the expression for the Bohr radius that everyone remembers

$$\Delta E = A\hbar^2 = \hbar^2 \frac{2e}{2mc} \frac{5.6e}{2Mc} \frac{1}{a_0^2} \frac{me^2}{\hbar^2} = \frac{\hbar^2}{2ma_0^2} \frac{5.6m}{M} \frac{e^4}{\hbar^2 c^2} = \frac{5.6m}{M} \alpha^2 Ry.$$

Substituting numbers roughly and using $hc = 1240 eV \cdot nm$ I got about $56cm$. The real splitting corresponds to $21cm$. For a rough estimate this is fine. A more careful calculation yields an extra factor of $8/3$ in the energy and yields a wavelength $3/8$ smaller, $21cm$.

At room temperature $\frac{P_{triplet}}{P_{singlet}} = 3e^{-(E_t - E_s)/(k_B T)} \approx 3$ where the factor of 3 comes from the degeneracy of the triplet and elementary counting in statistical mechanics. Please evaluate it at $2.7K$ using the correct value of $21cm$ for the splitting. Is it very different from room temperature?

Please try and explain simply why the singlet lies lower in energy.

4.5 You can do this in many ways.

$$\vec{S}_1 \cdot \vec{S}_2 / \hbar^2 = \frac{1}{4} \vec{\sigma}_1 \cdot \vec{\sigma}_2.$$

Thus

$$\mathcal{P}_1 = \frac{1}{4} (3 + \vec{\sigma}_1 \cdot \vec{\sigma}_2).$$

Note that we need

$$\frac{1}{16} (3I + \vec{\sigma}_1 \cdot \vec{\sigma}_2) (3I + \vec{\sigma}_1 \cdot \vec{\sigma}_2) = \frac{1}{16} (9I + 6\vec{\sigma}_1 \cdot \vec{\sigma}_2 + [\vec{\sigma}_1 \cdot \vec{\sigma}_2]^2).$$

To evaluate the last term you can use ϵ_{ijk} or you can be bone-headed and expand the terms:

$$(\sigma_{1x}\sigma_{2x} + \sigma_{1y}\sigma_{2y} + \sigma_{1z}\sigma_{2z}) (\sigma_{1x}\sigma_{2x} + \sigma_{1y}\sigma_{2y} + \sigma_{1z}\sigma_{2z}).$$

The “diagonal” terms yield the identity since $\sigma_{1x}^2 = I$.¹ There are 6 cross terms; we can evaluate one or more and write down the answer. $\sigma_{1x}\sigma_{2x}\sigma_{1y}\sigma_{2y} = -\sigma_{1z}\sigma_{2z} = \sigma_{1y}\sigma_{2y}\sigma_{1x}\sigma_{2x}$. The other terms yield the other components and so together we have $-2\vec{\sigma}_1 \cdot \vec{\sigma}_2$. Combining them together we obtain

$$\frac{1}{16} (9I + 6\vec{\sigma}_1 \cdot \vec{\sigma}_2 + [3I - 2\vec{\sigma}_1 \cdot \vec{\sigma}_2]) = \frac{3}{4}I + \frac{1}{4}\vec{\sigma}_1 \cdot \vec{\sigma}_2.$$

¹We are being cavalier here: $\sigma_{1x}^2 = I$ in spin-1 space and $\sigma_{2x}^2 = I$ in spin-2 space and $\sigma_{1x}^2 \sigma_{2x}^2 = I$ in the four-dimensional direct product space.

Since we have

$$(\vec{\sigma}_1 \cdot \vec{\sigma}_2)^2 = 3I - 2\vec{\sigma}_1 \cdot \vec{\sigma}_2$$

the other identity follows trivially.

Since $\vec{S} \cdot \vec{S}$ commutes with $\vec{S}_1 \cdot \vec{S}_2$ the projection operators $\mathcal{P}_1$ and $\mathcal{P}_0$ commute with $\vec{S} \cdot \vec{S}$ and they have simultaneous eigenfunctions.

We know from the previous problem that $\vec{S}_1 \cdot \vec{S}_2 / \hbar^2$ has eigenvalue $1/4$ for the triplet and $-3/4$ for the singlet. We saw this in the lecture while discussing Heisenberg exchange. Thus $\mathcal{P}_1$ has eigenvalue 1 acting on the triplet subspace and acting 0 on the singlet subspace and vice versa for $\mathcal{P}_0$. This establishes the required result.